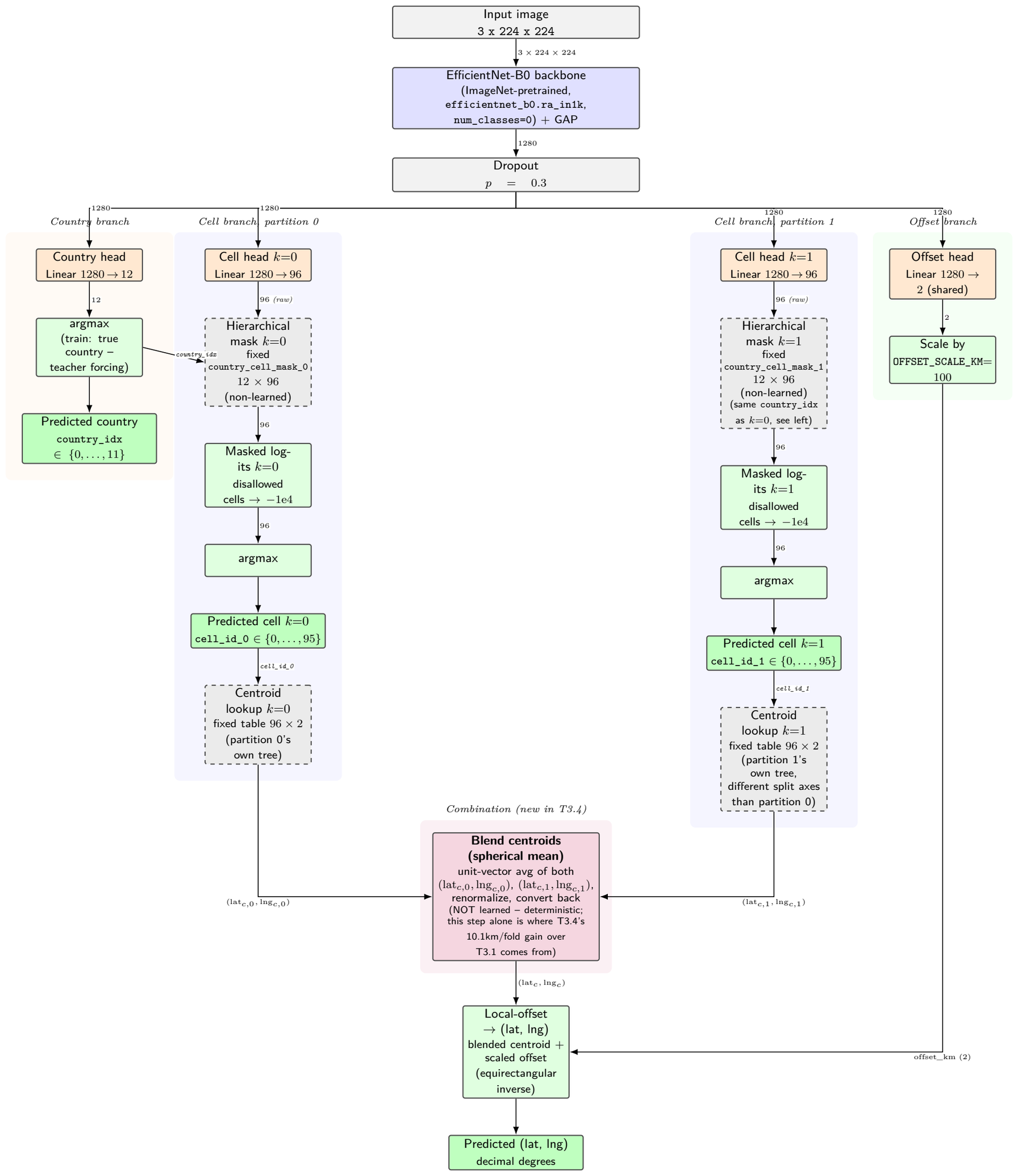
Input image
3 x 224 x 224
3 × 224 × 224
EfficientNet-B0 backbone
(ImageNet-pretrained,
efficientnet_b0.ra_in1k,
num_classes=0) + GAP
1280
Dropout
p = 0.3
1280
Country branch
Cell branch, partition 0
Cell branch, partition 1
Offset branch
Country head
Linear 1280 → 12
12
argmax
(train: true
country –
teacher forcing)
Predicted country
country_idx
∈ {0, . . . , 11}
country_idx
Cell head k=0
Linear 1280 → 96
96 (raw)
Hierarchical
mask k=0
fixed
country_cell_mask_0
12 × 96
(non-learned)
96
Masked log-
its k=0
disallowed
cells → −1e4
96
argmax
Predicted cell k=0
cell_id_0 ∈ {0, . . . , 95}
cell_id_0
Centroid
lookup k=0
fixed table 96 × 2
(partition 0's
own tree)
Cell head k=1
Linear 1280 → 96
96 (raw)
Hierarchical
mask k=1
fixed
country_cell_mask_1
12 × 96
(non-learned)
(same country_idx
as k=0, see left)
96
Masked log-
its k=1
disallowed
cells → −1e4
96
argmax
Predicted cell k=1
cell_id_1 ∈ {0, . . . , 95}
cell_id_1
Centroid
lookup k=1
fixed table 96 × 2
(partition 1's
own tree,
different split axes
than partition 0)
Offset head
Linear 1280 →
2 (shared)
2
Scale by
OFFSET_SCALE_KM=
100
Combination (new in T3.4)
Blend centroids
(spherical mean)
unit-vector avg of both
(lat_c,0, lng_c,0), (lat_c,1, lng_c,1),
renormalize, convert back
(NOT learned – deterministic;
this step alone is where T3.4's
10.1km/fold gain over
T3.1 comes from)
(lat_c,0, lng_c,0)
(lat_c,1, lng_c,1)
(lat_c, lng_c)
Local-offset
→ (lat, lng)
blended centroid +
scaled offset
(equirectangular
inverse)
offset_km (2)
Predicted (lat, lng)
decimal degrees